

Adham Ehab Erfan Mahmoud

Wireless Communications & RF Engineer | Embedded Systems & Automation

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Professional Summary

Electronics & Communication Engineering graduate with hands-on experience in wireless system design, RF/embedded hardware, and AI-driven signal processing. Internship at **Siemens Egypt** covering AVR/ARM microcontrollers, serial protocols (UART, SPI, I²C), and ESP32-based IoT data acquisition. Graduation project delivered a deep-learning beamforming framework for multi-cell 5G MIMO networks using ray-traced channel simulations. Seeking to apply physical-layer expertise and embedded programming skills to RF engineering, wireless network deployment, and industrial automation roles.

Core Competencies

- MIMO & Antenna Systems
- 5G NR Physical Layer (SSB/CSI-RS)
- RF Link Budget Analysis
- OFDM Transceiver Design
- Channel Estimation & Equalisation
- Beamforming & Codebook Design
- Digital Signal Processing
- Spectrum & Propagation Analysis
- ARM / AVR Microcontrollers
- UART / SPI / I²C Protocols
- MATLAB / Simulink
- Python (Tensorflow, NumPy, Sionna)
- C / C++ Embedded Programming
- Git, LaTeX

Industry Experience

Embedded Systems Intern

Siemens Egypt

Summer 2024

Egypt

- Worked with AVR and ARM Cortex-M microcontrollers: startup code, memory mapping, and low-level peripheral driver development.
- Implemented and debugged serial protocols (UART, SPI, I²C) for sensor interfacing and inter-device communication.
- Programmed ESP32 for real-time data acquisition and wireless cloud streaming for remote industrial monitoring.
- Practiced schematic reading, hardware–software integration, and systematic debugging with oscilloscope and logic analyser.

AI Model Developer

Outlier / Aligner

2024 – 2025

Remote

- Developed and evaluated AI-generated outputs in mathematics and algorithm design; provided structured feedback to guide training pipelines.

R&D Volunteer

University Centre for Career Development (UCCD)

2024 – Present

Alexandria, Egypt

- Contributed to technical research documentation and engineering analysis reports.

Education

B.Sc. Electronics & Communication Engineering

Alexandria University — Faculty of Engineering

Expected June 2026

Alexandria, Egypt

- GPA: **3.56 / 4.0 (Excellent)** Graduation project: Neural Codebook Design for MIMO Beam Management
- Relevant coursework: Digital Communications, Wireless Networks, Antenna Theory, DSP, Control Systems, Embedded Systems, Machine Learning

Key Technical Projects

- 5G MIMO Neural Beamforming Framework — Wireless Communications / Deep Learning
- Designed and trained a deep-learning model (PyTorch + Sionna) to generate SSB and CSI-RS codebooks for multi-cell 5G, maximising spectral efficiency under strict feedback overhead constraints.
 - Simulated ray-traced multi-cell RF environments with overlapping BS coverage; modelled realistic multi-path propagation and inter-cell interference.
 - Built a unified end-to-end pipeline: initial access, beam refinement, CSI feedback compression, and predictive beam tracking.
 - Evaluated power allocation and interference trade-offs; *manuscript in preparation* for IEEE submission (2026).
- OFDM & 2x2 MIMO Physical-Layer Simulation — RF / Digital Communications
- Implemented a full OFDM transceiver in MATLAB: pilot-based channel estimation, MMSE equalisation, and cyclic-prefix management.
 - Extended to 2x2 MIMO with spatial multiplexing; measured BER across AWGN and Rayleigh fading channels over a wide SNR range.
- DQN-Based Frequency Jammer — Physical Layer Security / Spectrum Engineering
- Trained a Deep Q-Network agent to adaptively target FHSS frequency bands with no prior knowledge of the hopping pattern; implemented experience replay, target networks, and ϵ -greedy exploration.
- Reconfigurable Intelligent Surfaces (RIS) for 5G/6G — RF / Antenna Systems
- Analysed RIS phase-shift design strategies: compared alternating optimisation (AO) vs. deep-learning passive beamforming for SNR enhancement and coverage extension.

Technical Skills

RF & Wireless	MIMO, OFDM, 5G NR (SSB/CSI-RS), Beamforming, Link Budget Analysis, Antenna Design, Propagation Modelling, Spectrum Analysis
MATLAB	Digital Comms Toolbox, DSP Toolbox, RF Toolbox, Simulink, Beamforming & Channel Simulation
Python	NumPy, PyTorch, TensorFlow, Sionna, MATLAB
Embedded / HW	C, C++, ARM Cortex-M, AVR, ESP32, UART/SPI/I ² C, PCB Design, IoT, RTOS basics, ROS
AI / ML	Deep Learning, CNNs, Autoencoders, DQN, Channel Estimation, Adaptive Algorithms
Tools	Git, L ^A T _E X, MATLAB Simulink, Sionna, KiCad
Languages	Arabic (Native) English — Professional (IELTS Academic 6.0)

Certifications

IELTS Academic — Band 6.0 (British Council)	Deep Learning — NVIDIA
Embedded Systems Essentials — Siemens	Machine Learning — Generative Technology Center